



Soil and Water Contamination by Perfluoroalkyl Acids (PFAAs) at Fire Outbreak Sites in Ghana: Distribution and Screening-level Risk Assessment

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Abstract Per- and polyfluoroalkyl substances (PFAS) are persistent environmental contaminants of concern because of their mobility, persistence, bio-accumulation potential, and possible adverse health effects. In this study, only two legacy perfluoroalkyl acids (PFAAs) that include perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) were quantified. Fire-outbreak locations are potential points of release because firefighting foams, thermally affected PFAS-containing materials, industrial activities, wastewater, landfill leachate, and atmospheric deposition can contribute target PFAAs to surrounding soil and water. To our knowledge, this study provides the first field investigation of PFOA and PFOS at documented fire-outbreak sites in Africa.

Across 18 fire-affected soil sites, PFOA and PFOS ranged from non-detectable to 5.72 ng/g dw and non-detectable to 10.80 ng/g dw, respectively; lower-bound site means (non-detects assigned zero) were 1.12 and 1.03 ng/g dw. Among seven surface-water sites, PFOA was detected at 5/7 sites and PFOS at 4/7 sites, with ranges of non-detectable–0.793 ng/L and non-detectable–13.50 ng/L and lower-bound means of 0.357 and 2.17 ng/L, respectively. One shallow hand-dug well (WR 3W), interpreted separately from surface water, contained 5.14 ng/L PFOA and non-detectable PFOS. Using updated U.S. EPA 2024 oral reference doses in a deterministic screening model, Hazard Index (HI) exceeded 1 at one surface-water site (GR 6W, driven by PFOS) and at WR 3W (driven by PFOA), while other surface-water HIs were below 1. These Hazard Quotient (HQ)/HI values are screening indicators rather than predictions of disease and depend strongly on intake, body weight, exposure

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duration, treatment, and benchmark selection. Comparison with the U.S. EPA 4.0 ng/L drinking-water Maximum Contaminant Levels (MCLs) is contextual rather than a compliance determination because most samples were untreated environmental waters. The EU 0.10 µg/L 'Sum of PFAS' parametric value cannot be fully assessed because only two of the 20 listed PFAS were measured, and WHO has not finalized PFOA/PFOS guideline values. The findings therefore provide preliminary baseline evidence of spatially heterogeneous PFOA/PFOS occurrence and support expanded analyte suites, seasonal and groundwater monitoring, source apportionment, biomonitoring, and Ghana-specific exposure assessment.

Keywords Perfluoroalkyl acids · PFOA · PFOS · Soil · Surface water · Shallow groundwater · Fire-outbreak sites · Aqueous film-forming foam · Screening-level risk

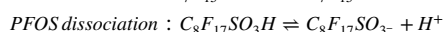
1 Introduction

Per- and polyfluoroalkyl substances (PFAS) comprise a broad class of synthetic organofluorine compounds used in industrial and consumer applications. The highly fluorinated carbon backbone contains strong C–F bonds that confer exceptional thermal and chemical stability and contribute to environmental persistence and long-range transport (Kukrálová et al., 2024a, b; Panieri et al., 2022). The present investigation did not quantify the full PFAS class; it specifically measured two legacy perfluoroalkyl acids (PFAAs), PFOA and PFOS.

Within PFAAs, perfluoroalkyl carboxylic acids (PFCAs) include perfluorooctanoic acid (PFOA; $C_8HF_{15}O_2$), whereas perfluoroalkane sulfonic acids (PFSAs) include perfluorooctane sulfonic acid (PFOS; $C_8HF_{17}O_3S$). Their anionic forms are shown in Fig. 1A. PFOA and PFOS were prioritized because they are well-studied legacy PFAAs with established environmental persistence and toxicological relevance (Kukrálová et al., 2024a, b).

Because only PFOA and PFOS were targeted, the measured concentrations represent target-PFAA occurrence rather than total PFAS burden. Consequently, PFOA/PFOS ratios are not treated as diagnostic source fingerprints and cannot distinguish AFFF inputs from landfill leachate, wastewater,

industrial discharges, atmospheric deposition, or PFAS-treated consumer/textile wastes. Modern firefighting foams may contain short-chain PFAS, fluorotelomer sulfonates (e.g., 6:2 FTS), ether PFAS, and precursor compounds that were not measured here. Discussion of these compounds is therefore used only to explain plausible environmental processes and future analytical needs, not as measured evidence in the present dataset. The physicochemical properties of the two measured PFAAs are summarized in Table 1.



At typical environmental pH, both compounds occur predominantly as anions. Their environmental partitioning can be summarized by $K_d = C_s/C_w$ and $K_{oc} = K_d/f_{oc}$, where C_s and C_w are sorbed and aqueous concentrations and f_{oc} is the soil organic-carbon fraction. A simplified retardation factor is $R = 1 + (\rho_b K_d/\theta)$, where ρ_b is soil bulk density and θ is volumetric water content. These relationships explain why sorption, soil organic carbon, texture, and hydrology are important covariates; these site-specific parameters were not measured in this study.

Fluorosurfactant-based firefighting foams, particularly aqueous film-forming foams (AFFF), are recognized sources of PFAS at airports, military installations, fuel depots, and fire-training areas (Ateş et al., 2025; Leeson et al., 2021; McGarr et al., 2023; Yan et al., 2024). However, AFFF-associated contamination normally contains mixtures that extend well beyond PFOA and PFOS. Accordingly, the present measurements should be interpreted as a lower-bound indicator of legacy PFAA occurrence and not as a complete characterization of foam chemistry or mixture risk.

African evidence remains comparatively sparse but demonstrates that PFAS can occur in drinking-water sources, wastewater-impacted waters, sediments, and biota. Studies from Uganda and South Africa, together with regional reviews, indicate contributions from wastewater, waste handling, industrial activities, and diffuse transport pathways (Ari-naitwe et al., 2021; Kasambala et al., 2024; Ojemaye et al., 2024). This limited monitoring base makes source-specific interpretation especially uncertain and underscores the value of carefully qualified baseline data.

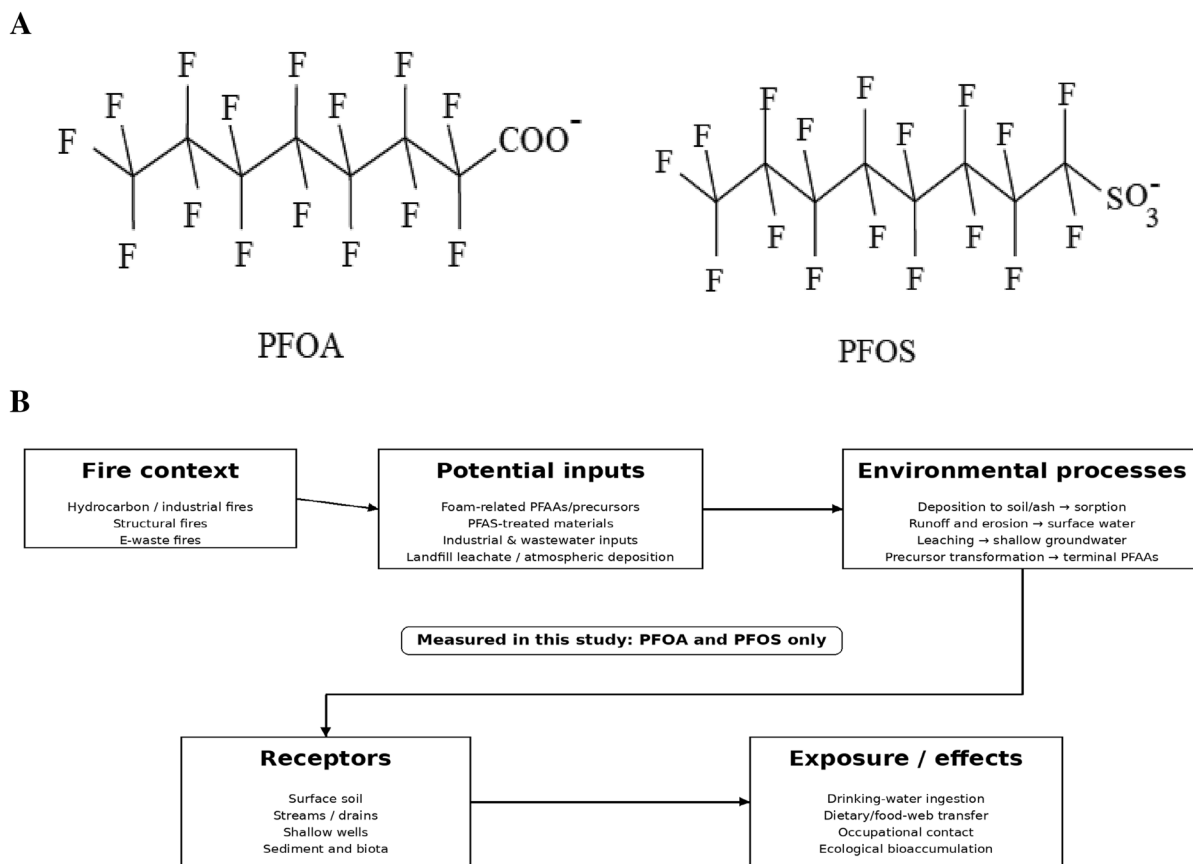


Fig. 1 A. Structural formulae of the anionic forms of PFOA and PFOS. B. Conceptual pathways linking fire context, potential PFAA inputs, environmental transport, receptors, and

exposure pathways. Only PFOA and PFOS were measured in this study; the other PFAS pathways shown are conceptual and were not analytically resolved

Table 1 Some physicochemical properties of PFOA and PFOS relevant to environmental fate

Property	PFOA	PFOS	Reference(s)
Chemical formula	C ₈ HF ₁₅ O ₂	C ₈ HF ₁₇ O ₃ S	Buck et al. 2011
Molecular weight (g/mol)	414.07	500.13	Kissa 2020
Water solubility at 25 °C	9.5 g/L	0.57 g/L	Ahrens 2011; OECD 2018
Vapor pressure (Pa at 25 °C)	3.0 × 10 ⁻⁴	3.3 × 10 ⁻⁴	USEPA 2023
Log Kow (estimated)	4.5	4.7	ATSDR 2021
pKa	2.8	<0	ATSDR 2021
Log Koc	2.3–2.8	2.7–3.3	Kukrálová et al. 2024a, 2024b
Henry's Constant (atm·m ³ /mol)	1.7 × 10 ⁻⁶	3.1 × 10 ⁻⁷	USEPA 2023

Potential PFOA/PFOS contributors in African urban environments include industrial discharges, municipal wastewater, landfill leachate, disposal of PFAS-containing consumer products, and atmospheric emissions followed by dry or wet deposition (Schroeder et al., 2021; Ssebugere et al., 2020;

Tang, 2025; Verma et al., 2023). These pathways can coexist with fire-related releases; therefore, no single source is assumed a priori in the present study.

Ghana experiences frequent structural, market, industrial, hydrocarbon, and e-waste fires. Different fire types can plausibly create different target-PFAA

pathways: hydrocarbon/industrial fires may involve foam application; structural fires can mobilize fluorinated treatments from textiles, furnishings, coatings, and building materials; and e-waste fires can release or redistribute fluorinated polymers and additives. Post-fire wash-off, ash deposition, runoff, leaching, wastewater inputs, and atmospheric deposition may subsequently redistribute PFOA and PFOS. These pathways are summarized conceptually in Fig. 1B.

Publicly available trade information supports the presence of firefighting-foam products in Ghana, but product brand, formulation, deployment volume, and incident-specific PFAS chemistry were not available for the sampled events. Accordingly, AFFF is treated as one plausible contributor rather than the dominant or unique source. Alternative contributors—including landfill leachate, wastewater, industrial discharges, PFAS-containing consumer/textile wastes, e-waste, and atmospheric deposition—are considered with equal caution.

Empirical information from West African fire-impacted environments remains limited. To our knowledge, this study is the first field investigation in Africa specifically examining PFOA and PFOS in soil and nearby waters at documented fire-outbreak sites. The objective was to establish preliminary baseline occurrence, describe spatial heterogeneity, examine PFOA/PFOS co-occurrence, and conduct screening-level exposure and risk interpretation. Because only two legacy PFAAs were quantified and the sampling design was cross-sectional, the study does not claim complete PFAS source apportionment, representative regional prevalence, or demonstrated soil-to-water transport mechanisms.

2 Materials and Methods

2.1 Sampling Site Identification

Sampling locations were selected from Ghana National Fire Service (GNFS) records to represent documented fire-outbreak settings in the Greater Accra, Western, and Ashanti Regions where foam use was plausible but could not be confirmed at the product or formulation level. Soil and nearby water receptors within approximately 500 m of the fire locations were sampled to characterize target-PFAA

occurrence. Site descriptions, fire context, land cover, and coordinates are provided in Table S1.

Six reference/control sites (two per region) were selected approximately 15 km from the fire-affected locations. Reference locations had no documented fire incident, major industrial activity, or known point source of PFOA/PFOS and were chosen within the same broad regional environmental setting to provide contextual background conditions. Because site-specific soil organic carbon, texture, and hydrogeological properties were not measured, comparability of reference and fire-affected locations is interpreted at a broad regional rather than mechanistic level.

2.2 Study Area

Ghana has a diverse geography and a dynamic economy driven by agriculture, mining, manufacturing, and services. Three key regions (Accra, Western, and Ashanti) selected for the study played significant roles in economic development. Accra, Ghana's economic hub, thrives in trade, manufacturing, real estate, and tourism. The Tema Industrial Area serves as a major manufacturing and export processing zone, while rapid urbanization fuels infrastructure growth. The Western Region is a center for oil, gas, and mining, with Takoradi serving as the logistics hub for offshore exploration. The region also supports agriculture, timber, and fishing, with the Takoradi Port facilitating maritime trade. The Ashanti Region is a powerhouse in gold mining, hosting some of Africa's largest mines. Cocoa farming is another key industry, alongside textile production, including the famous Kente cloth and handicrafts in Kumasi. These regions collectively contribute to Ghana's industrial and economic landscape. The map of Ghana showing the various sampling sites across the three regions is as shown in Fig. 2.

2.3 Sample Size and Collection

Sampling was conducted during a single campaign from October to December 2022 under broadly comparable seasonal conditions across the three study regions. The design was intended for baseline screening and spatial contrast rather than temporal trend analysis. Rainfall occurring immediately before each individual sampling event was not systematically recorded; therefore, event-scale runoff effects cannot

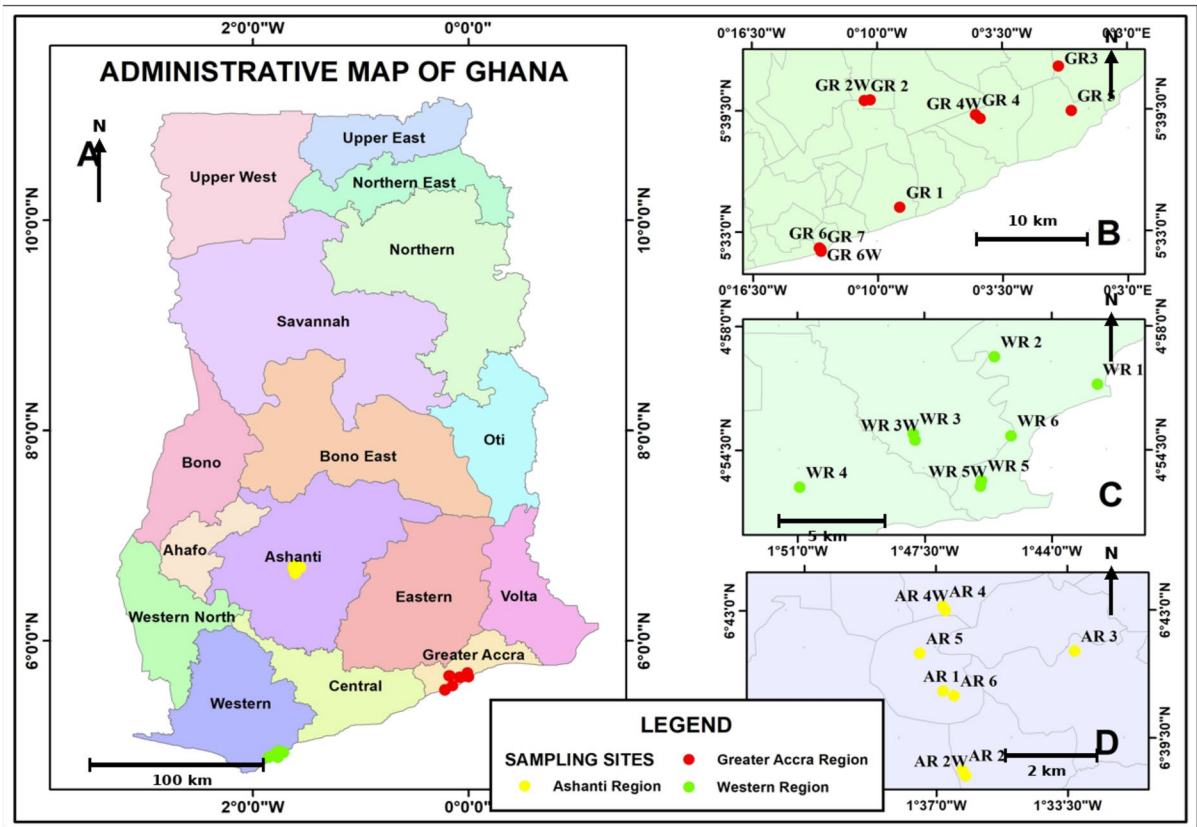


Fig. 2 A Administrative map of Ghana showing the three study regions; (B) Greater Accra sampling sites; (C) Western Region sampling sites; and (D) Ashanti Region sampling sites.

be separated from other hydrological controls. This is important because surface-water PFAA concentrations can change markedly after rainfall and with elapsed time since a fire incident (Aly et al., 2020).

The study was an exploratory baseline survey conducted under access and resource constraints. The achieved sample sizes were evaluated post hoc for mean-estimate precision rather than presented as a prospective statistical-power calculation. The precision relationship was $n = (Z_{1-\alpha/2} S_r / E)^2$, where $Z_{1-\alpha/2} = 1.96$ for 95% confidence, $S_r = IQR / 1.349$ is a robust dispersion estimate for skewed concentration data, and E is the absolute margin of error. Using the observed site-level dispersion, the achieved design corresponded approximately to 24% relative precision for soil ($n = 18$) and 19% for the combined water dataset ($n = 8$). These retrospective precision estimates do not remove the low power of regional comparisons, particularly for surface water.

North arrows and scale bars have been added to each map panel; regional panels are shown at different scales

The analytical dataset comprised 18 composite soil samples, seven surface-water samples, and one shallow hand-dug well (WR3W), collected from the three regions during October–December 2022. Soil comprised six sites per region. Surface water comprised four Greater Accra sites, two Ashanti sites, and one Western site (WR5W). WR3W was classified and interpreted separately as shallow groundwater because its transport processes differ from those of streams, drains, and gutters (Table 2).

Soil samples were extracted at a depth of 0–15 cm using a bucket auger. The bucket auger was thoroughly cleaned with methanol after each use to prevent cross-contamination. Five sub-samples that spread across the centre and four corners of a 5×5 m² area were mixed to form a composite sample. Approximately 1.0 kg of soil was collected and sealed in clean 500 mL zip-lock polypropylene (PP) bags. For control purposes, field blanks were mixed into the

Table 2 Summary of the sampling campaign and analytical matrices

Matrix/group	n	Regional distribution	Sampling period	Key interpretation
Soil	18	6 Greater Accra; 6 Ashanti; 6 Western	Oct–Dec 2022	0–15 cm composites; five subsamples per 5 × 5 m area
Surface water	7	4 Greater Accra; 2 Ashanti; 1 Western	Oct–Dec 2022	Within ~ 500 m; downstream receptor; ~ 10 cm below surface
Shallow groundwater	1	WR 3 W, Western Region	Oct–Dec 2022	Hand-dug well; interpreted separately from surface water
Reference/control sites	6	2 per region	Oct–Dec 2022	~ 15 km from fire sites; no documented fire/major industrial source

Rainfall in the days immediately preceding individual sampling events was not systematically recorded; therefore, event-specific rainfall was not used as a covariate

batch and subjected to the same environmental factors as the real samples. Following the removal of large stones and plant and animal remains, the soil samples were brought to the lab, air-dried, crushed, and homogenised using a silica mortar and pestle. Following screening via a 2 mm mesh, the samples were kept at room temperature in previously cleaned glass jars pending additional examination.

Surface-water samples were collected within approximately 500 m of the fire-outbreak location along the downstream watercourse. Pre-cleaned PFAS-compatible sampling bottles were rinsed with site water immediately before collection and filled at approximately 10 cm below the water surface. Samples were transported in a cooler box and refrigerated pending extraction. Surface-water results are treated as screening-level observations because concentrations may vary with rainfall, runoff, discharge, and time since the fire. WR 3 W, a shallow hand-dug well near a fuel-market fire site, was treated separately as a groundwater receptor.

2.4 Chemicals and Reagents

Labeled PFOA ($[^{13}\text{C}_8, 99\%]$) and PFOS ($[^{13}\text{C}_8, 99\%]$) were sourced from Cambridge Isotope Laboratories, USA, while PFOA (98%) and PFOS (98%) were acquired from ACCU Standard, USA. Ultra-pure methanol (Romil-UpSTM) and formic acid (99+%) were procured from Microsep and Thermo Scientific, South Africa, respectively. Ammonium hydroxide (25%) was acquired from Sigma-Aldrich and employed without further purification. Milli-Q system deionised water (resistivity > 18.5 M Ω -cm) was employed in making the working solutions.

2.5 AFFF Composition and its Importation by Ghana National Fire Service

Firefighting foams can contain complex PFAS mixtures, including short-chain PFAAs, fluorotelomer sulfonates (e.g., 6:2 FTS), and, in some formulations, ether PFAS (Ruyle et al., 2021; ITRC, 2022; USEPA, 2024d). These compounds are discussed to contextualize potential source chemistry only; they were not measured in this study. Trade records cannot identify the product used at a specific incident, and PFOA/PFOS occurrence alone is insufficient to establish an AFFF fingerprint.

GNFS product brand, safety data sheets, foam-generation chemistry, application volume, discharge duration, and incident-specific firefighting practice were not available for the sampled events. Therefore, the study does not assume that all fire sites experienced similar suppression practices, and differences among sites cannot be attributed quantitatively to foam volume or formulation. Future studies should integrate operational fire records with expanded target analysis, total oxidizable precursor (TOP) assays, and non-target/high-resolution screening.

2.5.1 Solid Phase Extraction

The soil sample extraction was done by the method outlined by Lorenzo et al. (2015). In summary, a combination of internal standards (I.S.), which is made up of 20 μL of 200 $\mu\text{g/L}$ each of isotopically labeled PFOA and PFOS, was added to all the samples to ensure uniformity and guarantee accuracy. Soil samples and internal standards were mixed for one minute using a vortex mixer. After that, 10 mL of

methanol was used to extract the soil samples in triplicates, centrifuged for 15 min at 965 rpm, vortexed, and then sonicated for another 15 min each time. The supernatant was brought to a final volume of 5 mL by passing nitrogen through it and ultra-pure water was used to adjust the volume to 100 mL. Following that, the reconstituted mixture was forced to flow at a rate of about 5 mL/min via a preconditioned Hypersep SPE (150 mg, 6 mL, WAX) cartridge. Nitrogen is made to pass through the cartridge for 20 min to ensure the removal of any remaining water. Following that, the cartridges were connected to an Envicarb system before elution, which required 5 mL of methanol with 0.1% NH_4OH . After that, the eluents were evaporated under nitrogen until the final volume was 1 mL. After clarifying the spiked extracts utilising a 0.22 μm syringe filter, they were moved to auto-sampler vials for additional examination.

2.5.2 Water Extraction

The refrigerated surface water samples were brought to room temperature prior to analysis. To ensure precision and reliability, a 20 μL mixture of internal standards, consisting of 200 $\mu\text{g/L}$ each of isotopically labelled PFOA and PFOS, was mixed with each of the samples to enhance accuracy and consistency. Using a vortex mixer, the mixture was homogenised for a minute. Then, using a method akin to soil extraction, 500 mL of ultra-pure water was pumped into the conditioned Hypersep SPE (150 mg, 6 mL, WAX) cartridge at a rate of 5 mL/min to precondition and elute it. Finally, nitrogen was pumped into the cartridge for 20 min to eliminate any remaining moisture from the sorbent.

The extraction process followed Taniyasu et al. (2005). Moderate anion exchange solid extraction was performed using Hypersep SPE (150 mg, 6 mL, WAX) cartridges. Before use, these cartridges were preconditioned by rinsing them with 18 M Ω deionised water after treatment with 5 mL of methanol with 0.1% NH_4OH . Water samples (500 mL) were pumped under vacuum via the C18 column cartridges at a flow rate of 5 mL per minute using polypropylene tubing. This extraction method successfully captured the target compounds. After extraction, all remaining inorganic ions and contaminants were removed from the cartridges by rinsing them with 18 M Ω deionised water. After

that, the cartridges were dried by centrifuging them for 15 min at 2500 rpm. Envicarb was connected to the cartridges, and 5 mL of methanol with 0.1% NH_4OH was passed through them for elution. Finally, the spiked extracts were evaporated under nitrogen to yield an overall amount of 0.5 mL. To guarantee clarity, the extracts were lastly passed through a 0.22 μm syringe filter before being moved to auto-sampler vials for further testing.

2.6 Liquid Chromatography–mass Spectrometry Analysis

An Agilent I-class System ultra-high-performance liquid chromatography (UPLC) and a Waters Xevo G2-XS Q-TOF mass spectrometer were used to measure the PFOA and PFOS concentration in the various environmental samples. At a flow rate of 0.3 mL/min, a mobile phase gradient of a solvent consisting of 45% acetonitrile and 55% water with 20 mM ammonium acetate was used to inject a sample volume of 1 μL into a BEH C18 column (ACQUITY UPLC, 1.7 μm , 2.1 $\times$ 50 mm) with a BEH C18 VanGuard Pre-Column (ACQUITY UPLC, 1.7 $\times$ 5 mm). In the negative mode, the Q-TOF MS was controlled by an electrospray ionisation (ESI) interface.

Representative negative-electrospray-ionization (ESI⁻) TOF–MS chromatograms from the laboratory analytical sequence are provided in Supplementary Figure S1: (A) a procedural blank, (B) the 10 ng/L calibration standard (instrument file Bernard_STDs_10_p), and (C) a representative environmental extract (instrument file Bernard_31_p). The images were cropped only to remove non-analytical desktop borders; the chromatographic traces, retention times, m/z annotations, and relative-intensity information were not altered. These chromatograms are used as qualitative QA/QC documentation. Quantitative concentrations remain based on isotope-dilution calibration, matrix recovery, blank correction, and matrix-specific LOD/LOQ criteria. For isotope-dilution calibration, the analyte-to-internal-standard response ratio is expressed as $R_i = A_i/AIS$, and concentration is obtained from $C_i = (R_i - b)/m$, where A_i is analyte peak area, AIS is internal-standard peak area, m is calibration slope, and b is the intercept.

2.7 Quality Assurance and Control

Quality assurance/quality control (QA/QC) included avoidance of PTFE-containing sampling and laboratory materials, methanol-cleaned equipment, field/procedural blanks, isotope-dilution internal standards, matrix spikes, calibration verification, blank correction, and replicate measurements. The ^{13}C -labelled PFOA and PFOS compounds served as isotope-dilution internal standards; a separate surrogate-compound set was not used. Thus, recovery statistics reported here refer to matrix-spike/method recovery of the target analytes rather than to independent surrogate recoveries.

Six-point calibration curves (0.1–50 ng/L) were used. Matrix-specific calibration performance was $R^2=0.997$ for PFOA and $R^2=0.998$ for PFOS in both soil and water validation sets. Triplicate precision was summarized as relative standard deviation, $\text{RSD}(\%)=(s/\bar{x})\times 100$. Independent field-duplicate pairs were not available; therefore, a separate duplicate RPD statistic is not reported. Blank-based sensitivity was calculated as $\text{LOD}=3s_{\text{blank}}$ and $\text{LOQ}=10s_{\text{blank}}$. Concentrations above the LOD but below the LOQ are retained only as estimated detections and are identified in Table 6.

Analytical uncertainty was interpreted conservatively from repeatability, recovery, calibration performance, and blank variability. Because the archived validation summary does not contain the full covariance structure, calibration slope/intercept uncertainty, or sufficient replicate levels required for a formal Guide to the Expression of Uncertainty in Measurement (GUM) budget, Table 5 is interpreted as a screening-level analytical-variability indicator rather than a formal expanded uncertainty estimate. This distinction prevents false precision in subsequent HQ/HI calculations.

2.8 Statistical Analysis

Statistical analyses were performed at the site level separately for soil and surface water; WR3W was excluded from surface-water regional tests. Non-detects were coded as zero for lower-bound descriptive statistics and screening-risk calculations, while detections between LOD and LOQ were retained as estimated values. Kruskal–Wallis tests evaluated differences in regional concentration distributions (not

means). Effect magnitude was summarized using epsilon-squared, $\varepsilon^2=(H-k+1)/(n-k)$, where H is the Kruskal–Wallis statistic, k is the number of groups, and n is the total number of observations. Where an omnibus test was significant, pairwise Mann–Whitney comparisons were evaluated with Holm correction for multiple testing. Median-centered Levene tests were used descriptively to examine heterogeneity of variance. Because n was small (especially surface water ($n=7$)) p -values were interpreted together with effect sizes, censoring frequency, and graphical patterns rather than as stand-alone evidence. Pearson PFOA/PFOS correlations were exploratory and were accompanied by 10,000-resample bootstrap 95% confidence intervals and leave-one-out influence checks.

2.9 Risk Analysis

Human-health evaluation was conducted as a deterministic screening exercise for direct ingestion of untreated water. It does not estimate clinical disease probability and does not include dietary, occupational, dermal, inhalation, or cumulative exposure to unmeasured PFAS. For analyte i , chronic daily intake (CDI), hazard quotient (HQ), and combined hazard index (HI) were calculated as follows:

$$CDI_i = (C_i \times IR \times EF \times ED) / (BW \times AT) \quad (1)$$

For chronic daily exposure, when $EF=365$ days/year and $AT=ED\times 365$ days, Eq. 1 reduces to $CDI_i=C_i\times IR/BW$.

$$HQ_i = CDI_i / RfD_i \quad (2)$$

$$HI = \sum HQ_i = HQ_{PFOA} = HQ_{PFOS} \quad (3)$$

where C_i is the measured concentration in water, IR is ingestion rate (1.6 L/day), EF is exposure frequency, ED is exposure duration, BW is body weight (60 kg), AT is averaging time, and RfD_i is the oral reference dose.

Updated U.S. EPA 2024 oral RfD s were used for the revised screening calculation: PFOA = 3×10^{-8} mg/kg-day (0.03 ng/kg-day) and PFOS = 1×10^{-7} mg/kg-day (0.10 ng/kg-day) (USEPA, 2024b, c).

Non-detects were assigned zero for a lower-bound screening estimate; this treatment is explicitly conservative with respect to reported concentration

burden but may understate exposure when true values lie between zero and the detection limit.

Benchmark comparison was kept separate from HQ/HI. The U.S. EPA 2024 enforceable drinking-water MCL is 4.0 ng/L for PFOA and 4.0 ng/L for PFOS (USEPA, 2024a). These values apply to regulated public drinking-water systems and are used here only as contextual screening benchmarks for environmental waters.

The EU Drinking Water Directive establishes 0.10 µg/L (100 ng/L) for the 'Sum of PFAS' and 0.50 µg/L (500 ng/L) for 'PFAS Total'; it does not establish a 20 ng/L individual PFOA or PFOS limit in Directive (EU) 2020/2184.

Because only PFOA and PFOS were measured, compliance with the EU 'Sum of PFAS' parameter cannot be determined. WHO has not finalized guideline values for PFOA/PFOS; its current advice is to minimize concentrations as low as reasonably practical while broader assessment continues (WHO, 2025).

The deterministic HQ/HI results are therefore interpreted as screening indicators subject to uncertainty in concentration, drinking-water use, treatment, exposure duration, body weight, sensitive life stages, and unmeasured co-exposures.

Probabilistic exposure analysis (e.g., Monte Carlo simulation using distributions for IR, BW, exposure frequency, and concentration) was beyond the available dataset and is recommended for follow-up risk assessment.

3 Results and Discussion

3.1 Quality Control and Assurance

Calibration was linear over 0.1–50 ng/L, with R^2 values of 0.997–0.998 for PFOA and PFOS across the

Table 4 Summary of the validation parameters for the analytical method used to detect and quantify PFOA and PFOS in the soil and surface water samples

A. Soil (ng/g dw)			
COMPOUND	R^2	RECOVERY, %	RSD, %
PFOA	0.997	83	14.85
PFOS	0.998	89	15.80
B. Water (ng/L)			
PFOA	0.997	89	16
PFOS	0.998	95	15

soil and water validation sets. Triplicate blank concentrations, matrix-specific blank means, SDs, LODs, and LOQs are reported in Table 3. Using $LOD = 3s_{\text{blank}}$ and $LOQ = 10s_{\text{blank}}$, soil/water LODs were 0.10 for PFOA and 0.19 for PFOS in the corresponding matrix units, and LOQs were 0.330 and 0.630, respectively. Values above LOD but below LOQ were treated as estimated detections rather than fully quantitative results.

Matrix-spike/method recoveries were 83% and 89% for PFOA and PFOS in soil and 89% and 95% in water, respectively, with triplicate RSDs of 14.85–16%. These values support acceptable screening-level trueness and repeatability for the two target analytes. Because separate surrogate compounds and independent field-duplicate pairs were not used, no surrogate-recovery or field-duplicate RPD values are claimed. Summary of the validation parameters for the analytical method used to detect and quantify PFOA and PFOS in the soil and surface water samples are shown in Table 4.

Table 5 summarizes screening-level analytical variability indicators derived from the available recovery, repeatability, calibration, and blank information. The approximately 19–25% relative values should be interpreted as an approximate sensitivity

Table 3 Triplicate blank concentrations and blank-based sensitivity values

Matrix	Analyte	Blank concentrations	Mean	SD	LoD	LoQ
Soil	PFOA	0.017, 0.050, 0.083	0.050	0.033	0.10	0.330
Soil	PFOS	0.037, 0.100, 0.163	0.100	0.063	0.19	0.630
Water	PFOA	0.027, 0.060, 0.093	0.060	0.033	0.10	0.330
Water	PFOS	0.057, 0.120, 0.183	0.120	0.063	0.19	0.630

Blank concentrations are expressed in the corresponding matrix units (ng/g for soil and ng/L for water). SD is the sample standard deviation of the three blank concentrations

envelope, not as a formal expanded measurement uncertainty. This analytical variability propagates approximately linearly into deterministic HQ values when exposure assumptions are held fixed.

A summary of site-level PFOA and PFOS concentrations in soil, surface water, and the single shallow well is presented in Table 6

The representative instrument chromatograms provide an additional qualitative check on the analytical sequence (Supplementary Figure S1). The procedural blank contains measurable late-eluting background ions, including signal near m/z 499, which is consistent with the decision to quantify and correct matrix-specific blanks rather than infer analyte presence from chromatographic appearance alone. By comparison, the 10 ng/L calibration standard shows prominent responses at approximately 15.22 min (m/z 369.0183) and 15.82 min (m/z 498.9867). The representative environmental extract shows corresponding ions at approximately 15.12 min (m/z 369.0123) and 15.66 min (m/z 498.9771). This retention-time/mass agreement is supportive of the PFOA/PFOS analytical assignment within the original Q-TOF workflow, but it is not treated as independent quantitative evidence. Because the archived presentation contains base-peak chromatograms rather than the complete raw-data integration reports, concentration estimates continue to rely on the validated isotope-dilution calibration, recovery, blank, LOD, and LOQ procedures described above.

3.2 Concentrations of PFOA and PFOS in the Soil

PFOA was detected in 15/18 soil sites (83.3%) and PFOS in 10/18 (55.6%). Site-level concentrations ranged from non-detectable to 5.72 ng/g dw for PFOA and non-detectable to 10.80 ng/g dw for PFOS. Using zero for non-detects as a lower-bound

convention, site means calculated from Table 6 were 1.12 ng/g dw (PFOA) and 1.03 ng/g dw (PFOS). For PFOA, the site-level regional distributions did not differ significantly (Kruskal–Wallis $H=3.91$, $p=0.142$, $\epsilon^2=0.127$), although median-centered Levene testing indicated heterogeneity of variance ($p<0.001$). For PFOS, the exploratory omnibus test indicated regional heterogeneity ($H=7.42$, $p=0.024$, $\epsilon^2=0.361$), but Holm-adjusted pairwise comparisons did not retain statistical significance (all adjusted $p\geq 0.062$). Thus, the prominent WR 2 value (10.8 ng/g) is best interpreted as a localized high observation rather than evidence that the Western Region as a whole is uniformly contaminated.

Greater Accra showed several low-to-moderate estimated PFOS detections and detectable PFOA, whereas Ashanti had fewer PFOS detections. The Western dataset included the largest individual soil values. These patterns are consistent with spatially heterogeneous target-PFAA inputs, but the small number of sites per region and absence of site-specific source tracers prevent attribution to a single regional economic activity. Landfill leachate, municipal waste, industrial discharge, atmospheric deposition, PFAS-containing materials, and firefighting-related inputs are all plausible contributors.

The WR 2 PFOS concentration warrants follow-up because PFOS is persistent, can accumulate in biota, and may transfer through terrestrial and aquatic food webs. However, this study did not measure soil biota, crop uptake, sediment, species-specific toxicity endpoints, or site-specific ecological benchmarks; therefore, the data do not support a formal conclusion that an ecological effect threshold was exceeded. Ecological significance is more appropriately framed as a monitoring priority requiring sediment/biota sampling, bioaccumulation assessment, and species-relevant risk benchmarks.

Table 5 Screening-level analytical variability indicators derived from reported QA/QC metrics

Matrix	Analyte	Blank SD	LoQ	u_blank	u_RSD	u_rec	u_cal	Combined u_rel (%)
Soil	PFOA	0.033	0.330	0.100	0.148	0.170	0.003	24.7
Soil	PFOS	0.063	0.630	0.100	0.158	0.110	0.002	21.7
Water	PFOA	0.033	0.330	0.100	0.160	0.110	0.003	21.8
Water	PFOS	0.063	0.630	0.100	0.150	0.050	0.002	18.7

Comparison with African studies should likewise be interpreted cautiously because studies differ in analyte suites, matrices, hydrology, analytical methods, and source context. The 2025 Pretoria study by Okwuosa et al. is confirmed as a published peer-reviewed article and demonstrates that wastewater and sediment partitioning can produce strong local variation in both legacy and emerging PFAS. The Ghanaian data therefore add a fire-site baseline to the broader African evidence base rather than establishing representative regional concentrations.

3.3 Surface Water and Shallow Groundwater Concentrations

Among the seven surface-water sites, PFOA was detected at 5/7 sites (71.4%) and PFOS at 4/7 sites (57.1%). PFOA ranged from non-detectable to 0.793 ng/L and PFOS from non-detectable to 13.50 ng/L. With non-detects assigned zero, lower-bound surface-water means were 0.357 ng/L for PFOA and 2.17 ng/L for PFOS. Regional testing was underpowered because the design comprised four Greater Accra, two Ashanti, and one Western surface-water site. Exploratory Kruskal–Wallis tests did not provide evidence of regional differences for PFOA ($H=4.43$, $p=0.109$) or PFOS ($H=4.85$, $p=0.089$). These values are therefore preliminary site observations rather than representative estimates for the three regions.

WR3W was the only shallow-groundwater receptor and contained 5.14 ng/L PFOA with PFOS non-detectable. It is excluded from the surface-water mean, detection frequency, correlation, and regional tests. Its elevated PFOA relative to the surface-water sites is consistent with, but does not prove, shallow subsurface transport. Site-specific groundwater gradient, depth-to-water, hydraulic conductivity, recharge, soil texture, and organic carbon were not measured, so the transport mechanism cannot be resolved.

Measured surface-water concentrations were lower than many values reported at heavily impacted AFFF training or industrial sites in North America, Europe, and Asia, but cross-study comparisons should account for different fire histories, sampling times, hydrology, analytical methods, and PFAS suites. Runoff, wastewater, landfill leachate, industrial discharge, atmospheric deposition, and contaminated soil can all contribute PFOA/PFOS

Table 6 PFOA and PFOS concentrations at selected fire-outbreak sites in Ghana (soil, seven surface-water sites, and one shallow hand-dug well)

Sampling Point	PFOA		PFOS	
	Avg. Concentration in soil (ng/g)	Std. Dev	Avg. Concentration in soil (ng/g)	Std. Dev
GR 1	nd	0	0.421†	0.013
GR 2	0.442	0.002	0.592†	0.005
GR 3	0.366	0.009	0.548†	0.009
GR 4	0.351	0.001	0.381†	0.002
GR 5	0.500	0.015	nd	0
GR 6	0.641	0.002	nd	0
AR 1	nd	0	nd	0
AR 2	0.447	0.005	nd	0
AR 3	1.30	0.015	nd	0
AR 4	0.471	0.003	nd	0
AR 5	0.444	0.003	nd	0
AR 6	0.344	0.002	0.381†	0.002
WR 1	1.54	0.011	0.960†	0.002
WR 2	4.04	0.026	10.8	0.007
WR 3	3.03	0.02	2.22	0.02
WR 4	0.462	0.002	0.381†	0.002
WR 5	nd	0	nd	0
WR 6	5.72	0.02	1.81	0.021
	Avg. Concentration in water (ng/L)	Std. Dev	Avg. Concentration in water (ng/L)	Std. Dev
GR 2W	nd	0	0.491†	0.002
GR 4W	0.411	0.006	0.381†	0.001
GR 6W	0.355	0.003	13.5	0.012
GR 7W	0.374	0.002	0.806	0.003
AR 2W	0.793	0.003	nd	0
AR 4W	0.563	0.003	nd	0
WR 3W	5.14	0.02	nd	0
WR 5W	nd	0	nd	0

nd=non-detect (<LOD). †=detected above the LOD but below the LOQ; these values are interpreted as estimated concentrations. WR 3W is shallow groundwater and is not included in surface-water summary statistics

to surface waters; consequently, the observed concentrations cannot be assigned uniquely to fire-fighting foam.

Alternative pathways are particularly relevant in urban and industrial catchments. Wastewater effluent, landfill leachate, PFAS-containing consumer

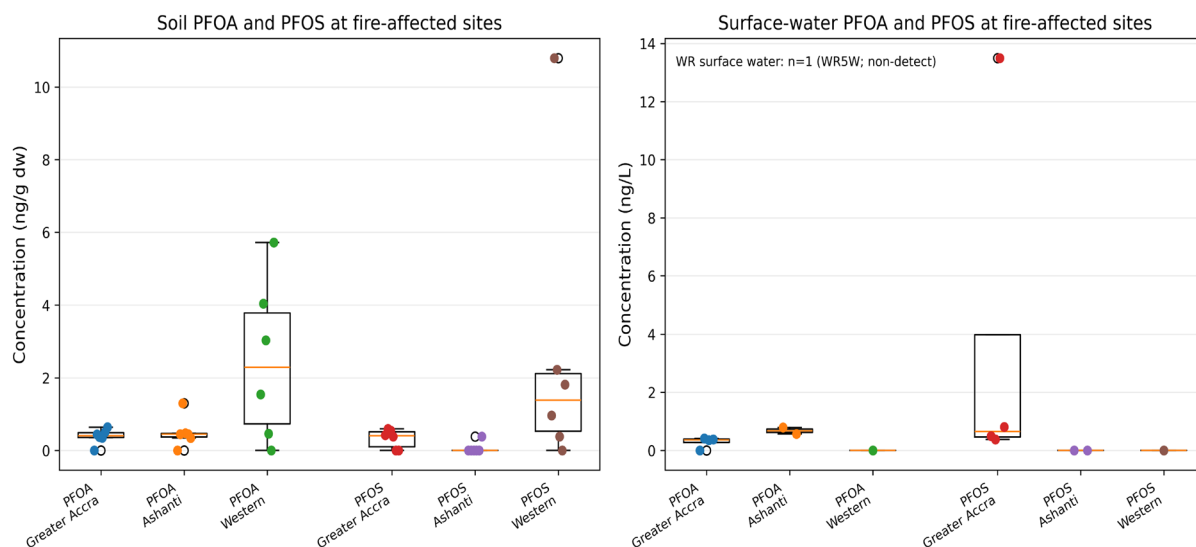


Fig. 3 A boxplot showing comparative PFAS distribution in soil and water samples from fire outbreaks sites in the three regions of Ghana

and textile wastes, industrial emissions, and atmospheric transport followed by wet/dry deposition can generate background or co-located inputs (Dalahmeh et al., 2018; Helmer et al., 2021; Schroeder et al., 2021; Tang, 2025). The single-campaign design cannot partition the relative contribution of these pathways.

Within the African literature, the Ghana measurements fall within the broad range reported for wastewater- and urban-impacted waters, although direct comparisons remain method- and site-dependent (Arinaitwe et al., 2021; Ojemaye et al., 2024; Okwuosa et al., 2025; Ssebugere et al., 2020). GR6W, located near a fire-training setting, contained the highest surface-water PFOS concentration (13.50 ng/L) and merits targeted repeat sampling with an expanded PFAS suite and source tracers.

3.4 Comparative Analysis of PFOA and PFOS among the Study Regions

Figure 3 summarizes site-level PFOA and PFOS distributions by region. The figure is intended as a descriptive baseline visualization. Regional inference is limited by six soil sites per region and particularly by the unbalanced surface-water design (Greater

Accra $n=4$, Ashanti $n=2$, Western $n=1$). WR3W is excluded from the surface-water panel.

The larger individual values in Western Region soil and the GR 6W PFOS surface-water value indicate spatial heterogeneity among sampled sites, not a region-wide contamination gradient. Because land use, source inventory, fire severity, foam formulation/volume, soil organic carbon, rainfall, and hydrology were not measured consistently, apparent spatial contrasts remain hypothesis-generating.

Retention and release can be interpreted using partitioning relationships. For a target PFAA, $K_d = C_s/C_w$ and $K_{oc} = K_d/f_{oc}$; stronger sorption generally favors soil/sediment retention, whereas lower sorption favors aqueous mobility. PFOS commonly exhibits greater sorption than PFOA, but site-specific K_d values cannot be calculated here because paired soil–water concentrations and soil organic-carbon fractions were not measured along defined flow paths.

The fire-outbreak setting remains environmentally relevant because it creates a plausible convergence of multiple target-PFAA inputs and hydrological redistribution pathways. Nevertheless, the present spatial patterns should be interpreted as preliminary baseline observations at selected sites rather than representative conditions for Greater Accra, Ashanti, or Western Region populations.

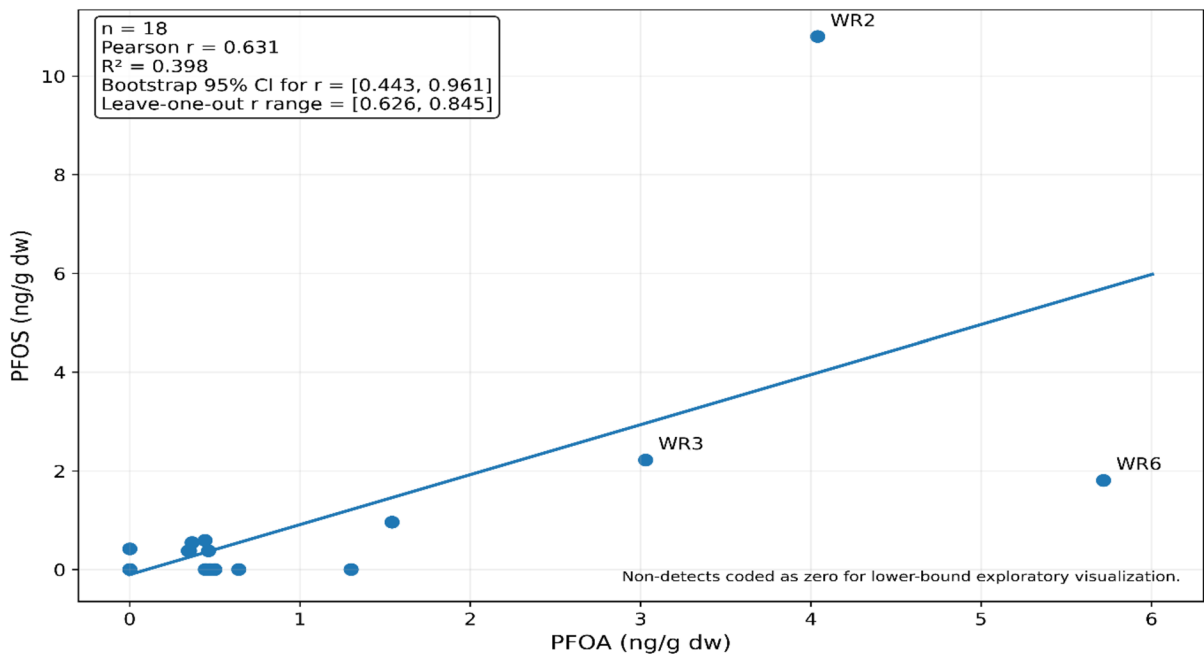


Fig. 4 Soil PFOA–PFOS scatter plot with site-level concentrations, fitted linear trend, bootstrap confidence interval for Pearson r , and leave-one-out influence range

The principal value of the regional comparison is therefore to identify locations for follow-up, not to rank whole regions. Future designs should use denser spatial coverage, paired source–soil–water transects, repeated wet/dry-season sampling, and measured covariates such as organic carbon, texture, discharge, groundwater gradient, rainfall, and time since fire.

3.5 Scatter Plots for PFOA and PFOS in Soil and Surface Water

Scatter plots were used only to visualize site-level PFOA/PFOS co-occurrence within each matrix. Pearson r is reported with a 10,000-resample bootstrap 95% confidence interval and leave-one-out sensitivity range. Because non-detects were coded as zero and the data contain high-leverage observations, the correlations are exploratory and are not interpreted as proof of shared source, causal coupling, or soil-to-water transfer. The Soil PFOA–PFOS scatter plot, showing site-level concentrations, is presented in Fig. 4.

Across 18 soil sites, PFOA and PFOS showed a moderate positive exploratory association (Pearson $r = 0.631$; $R^2 = 0.398$; bootstrap 95% CI for $r \approx$

0.446–0.960). The leave-one-out r range (approximately 0.626–0.845) indicates that the positive direction persists when individual observations are removed, although the magnitude is influenced by high-concentration Western sites. Because only two PFAAs were measured, this co-occurrence cannot identify a specific source.

For the seven surface-water sites, PFOA/PFOS co-variation was essentially absent (Pearson $r = -0.019$; $R^2 < 0.001$; bootstrap 95% CI approximately -0.861 to 0.570). The wide confidence interval reflects the very small sample size, censoring, and the influential GR 6W PFOS concentration. This result should therefore be described as statistically inconclusive rather than as evidence that the analytes are controlled by definitively different mechanisms.

WR 3W is shown separately in the text because it is shallow groundwater rather than surface water. Its PFOA concentration (5.14 ng/L) and non-detectable PFOS may reflect different subsurface retention and transport processes, but one well cannot establish a groundwater pattern.

Differential mobility remains a plausible literature-based explanation: PFOS generally sorbs more

Table 7 Deterministic screening-level HQ and HI values using U.S. EPA 2024 oral RfDs

Site	Matrix	PFOA (ng/L)	PFOS (ng/L)	HQ PFOA	HQ PFOS	HI
GR 2W	Surface water	0	0.491	0.00	0.13	0.13
GR 4W	Surface water	0.411	0.381	0.37	0.10	0.47
GR 6W	Surface water	0.355	13.5	0.32	3.60	3.92
GR 7W	Surface water	0.374	0.806	0.33	0.21	0.55
AR 2W	Surface water	0.793	0	0.70	0.00	0.70
AR 4W	Surface water	0.563	0	0.50	0.00	0.50
WR 3W	Shallow groundwater	5.14	0	4.57	0.00	4.57
WR 5W	Surface water	0	0	0.00	0.00	0.00

Assumptions: IR = 1.6 L/day; BW = 60 kg; chronic daily exposure; PFOA RfD = 0.03 ng/kg-day; PFOS RfD = 0.10 ng/kg-day; non-detects = 0. HQ/HI are screening ratios and do not represent disease probabilities

strongly to soil and sediment, while PFOA is often more mobile in water (Xing et al., 2023; Ackerman Grunfeld et al., 2024; Fatima et al., 2025). However, site-specific retardation would depend on $R = 1 + (\rho_b K_d / \theta)$; because ρ_b , K_d , organic carbon, θ , and groundwater flow were not measured, no mechanistic transport coefficient can be estimated from the present data.

Accordingly, the scatter plots are used as descriptive evidence of co-occurrence patterns only. Their uncertainty reinforces the need for expanded analyte panels, larger sample sizes, repeated sampling, and paired hydrological measurements before source or transport models are fitted.

3.6 Potential Exposure, Human-health Screening, and Ecological Considerations

The revised risk analysis separates surface water from the WR3W shallow well and uses the final U.S. EPA 2024 oral RfDs rather than the superseded 2022 interim toxicity values. Under the deterministic assumptions of IR = 1.6 L/day and BW = 60 kg, HI exceeded 1 at GR 6W (HI = 3.92; driven by PFOS) and at WR 3W (HI = 4.57; driven by PFOA). All other surface-water sites had HI < 1. These values are screening ratios, not estimates of individual disease probability.

Benchmark selection materially changes interpretation. The U.S. EPA 2024 enforceable MCLs are 4.0 ng/L for PFOA and PFOS in regulated public drinking-water systems. GR 6W PFOS (13.50 ng/L) and WR 3W PFOA (5.14 ng/L) are above those concentrations, whereas the remaining measured

surface-water concentrations are below 4.0 ng/L. Because these samples were environmental waters rather than compliance samples from regulated public-water systems, these comparisons are contextual screening indicators only.

The EU Drinking Water Directive uses 100 ng/L for the 'Sum of PFAS' (20 specified PFAS) and 500 ng/L for 'PFAS Total'. Because 18 of the 20 PFAS contributing to the EU sum were not measured, the study cannot determine EU compliance from PFOA and PFOS alone. WHO has not finalized numerical guideline values for PFOA/PFOS and currently advises Member States to reduce concentrations as low as reasonably practical while a broader assessment continues (WHO, 2025).

Exposure uncertainty is at least as important as analytical uncertainty. Actual intake depends on whether a sampled water body is used for drinking, treatment practices, intake rate, body weight, duration and frequency of use, and co-exposures from food, consumer products, and occupation. Children, pregnant people, highly exposed workers such as firefighters, and households relying on untreated shallow wells may have different exposure distributions from the single adult screening scenario used here. A probabilistic assessment would model these variables rather than fixing them at point values.

Ecological implications also warrant attention, but the present study did not measure sediment, biota, tissue burdens, or species-specific toxicity endpoints. PFOA and especially PFOS can persist and bioaccumulate, and food-web transfer has been documented in aquatic systems. Accordingly, the GR6W and high-soil observations identify locations for sediment,

aquatic-biota, crop, and food-web follow-up; they do not by themselves establish ecological harm.

Table 7 presents the revised lower-bound HQ/HI calculations. Non-detects were assigned zero. The values should be interpreted together with the approximately 19–25% analytical-variability envelope and the larger, unquantified uncertainty associated with real-world exposure behavior.

3.7 Policy and Public Health Implications

These findings support development of a Ghana-specific framework for target-PFAA/PFAS surveillance while recognizing that the present dataset is preliminary. Priority actions include targeted follow-up of GR6W and WR3W, national inventory of fire-fighting-foam products, documentation of incident-specific foam use, monitoring of landfill/wastewater/industrial and atmospheric pathways, and inclusion of PFAS in drinking-water source surveillance. International benchmarks can inform interim decision-making, but Ghana-specific standards should be based on local exposure patterns, analytical capacity, treatment feasibility, and protection of sensitive populations.

3.8 Potential Remediation Strategies

3.8.1 For Soil Remediation

Excavation and containment may be the most immediately feasible option for highly contaminated soils (e.g., > 100 ng/g PFOS) near fire training areas. Given Ghana's developing waste management infrastructure, excavated soils could be stored in lined, secure landfills to prevent leaching until more advanced treatment options become available. For less severely contaminated soils, in situ stabilization using activated carbon or other sorbents could reduce PFAS mobility, particularly in areas with sandy soils where leaching risk is high (Bolan et al., 2021). Pilot testing with locally available biochar (e.g., from coconut shells or agricultural waste) could provide a low-cost alternate surface water treatment.

Given the mobility of PFAS in water, pump-and-treat systems using granular activated carbon (GAC) or anion exchange resins could be implemented at key discharge points, such as drainage outflows near fire stations (Niarchos et al., 2021; Zeng et al., 2020). Modular, containerized treatment units could be

deployed at high-priority sites (e.g., near hand-dug wells) to provide interim protection while long-term solutions are developed. For smaller-scale or remote contamination, constructed wetlands with PFAS-accumulating plants (e.g., *Phragmites australis*) could serve as a passive, low-maintenance treatment option, though their effectiveness for complete PFAS destruction remains limited (Khan et al., 2023) alternative to commercial activated carbon (Bolan et al., 2021).

3.9 AFFF-Related PFAS Inputs, Persistence, and Risk Pathways

This study provides field evidence of PFOA and PFOS occurrence at Ghanaian fire-outbreak sites, but the analytical design cannot establish firefighting foam as the dominant or unique source. The measured pattern is compatible with several pathways, including fire-suppression inputs, release from burned PFAS-containing materials, industrial discharge, wastewater, landfill leachate, and atmospheric deposition. Future source attribution should combine expanded PFAS fingerprints, precursor assays, non-target screening, and incident-specific foam records.

Cross-media occurrence was observed in soils, surface waters, and one shallow well, but the study did not measure paired mass fluxes or hydraulic connectivity and therefore cannot demonstrate soil-to-water migration. Observed concentrations are consistent with hydrologically plausible redistribution by runoff and leaching. Definitive transport assessment would require paired transects, discharge/recharge data, groundwater gradients, soil organic carbon and texture, and repeated sampling through wet and dry periods.

The single October–December 2022 campaign represents one temporal snapshot. PFOA/PFOS concentrations can vary with rainfall, groundwater recharge, seasonal dilution, remobilization, and time elapsed since the fire. Consequently, persistence cannot be quantified from this cross-sectional dataset even though long environmental persistence of legacy PFAAs is well established in the literature.

The policy significance is therefore precautionary and monitoring-oriented: Ghana currently lacks a dedicated national PFOA/PFOS drinking-water or soil standard, while communities may encounter

multiple exposure routes. A coordinated program involving EPA Ghana, the Water Resources Commission, GNFS, water utilities, and research laboratories could prioritize high-risk sites, document foam inventories and disposal, expand analytical capacity, and generate the longitudinal data required for risk-based national standards.

4 Conclusion

This study provides preliminary baseline evidence of PFOA and PFOS occurrence in soil and nearby waters at selected Ghanaian fire-outbreak sites and, to our knowledge, represents the first field investigation of this type in Africa. The data show pronounced site-level heterogeneity, including high PFOS at WR 2 soil and GR 6W surface water and high PFOA in the WR 3W shallow well. These patterns are consistent with multiple potential sources and transport pathways; they do not establish AFFF as the unique source, demonstrate soil-to-water transfer, or represent region-wide contamination conditions. Using updated U.S. EPA 2024 RfDs, deterministic HI exceeded 1 at GR 6W and WR 3W, while other surface-water sites were below 1. The screening results remain sensitive to exposure assumptions and do not account for diet, occupation, sensitive populations, or unmeasured PFAS. For Ghana, the principal implication is the need for a coordinated monitoring and source-management framework rather than immediate extrapolation from a limited cross-sectional dataset.

- Develop national guidance for procurement, selection, use, containment, and disposal of firefighting foams, including an inventory and phase-out strategy for legacy PFAS-containing products where feasible.
- Establish a targeted PFAA/PFAS monitoring program led by EPA Ghana and the Water Resources Commission, prioritizing recurrent fire sites, drinking-water sources, shallow groundwater, wastewater/landfill pathways, and vulnerable communities.
- Apply confirmatory site assessment before remediation and use risk-based treatment options appropriate to local soil, groundwater, and water-use conditions.
- Require better incident records for foam type, quantity, duration of application, runoff containment, and post-fire cleanup so that future environmental data can be linked to operational practice.
- Build national analytical capacity for expanded PFAS suites, TOP assay, non-target/high-resolution screening, seasonal monitoring, groundwater monitoring, food/biota testing, and human biomonitoring.

By framing the results as a targeted PFOA/PFOS baseline, explicitly separating measured evidence from plausible mechanisms, and identifying the major data gaps, the study provides a defensible foundation for Ghanaian environmental surveillance and future risk-reduction policy.

4.1 Limitations of this Study

4.1.1 Absence of Groundwater Monitoring Data

The primary water design comprised seven surface-water sites and one shallow hand-dug well. WR3W is explicitly classified as shallow groundwater and interpreted separately. Because only one groundwater receptor was sampled, the study cannot characterize groundwater contamination or hydrogeological gradients. Future work should include monitoring wells, water-table depth, hydraulic conductivity, groundwater flow direction, and seasonal recharge.

4.1.2 Limited Surface-water Sample Size and Statistical Power

The surface-water dataset ($n=7$) is small and unbalanced among regions, which limits statistical power and makes estimates sensitive to individual observations such as GR6W. Regional comparisons are therefore exploratory. Future work should increase the number of sites per region, use repeated wet- and dry-season sampling, and pre-specify power/effect-size targets.

4.1.3 Limited Ability to Confirm Transport Mechanisms

The study was not designed to establish transport mechanisms. Site-specific soil organic carbon, texture, bulk density, porosity, discharge, runoff direction, distance-to-water, groundwater gradient, recharge, and time since fire were not measured.

Without these variables, K_d , K_{oc} , retardation factors, and contaminant mass flux cannot be estimated. Future paired source–soil–water transects should explicitly measure these hydrogeological and soil controls.

4.1.4 Limited Source Apportionment

Only PFOA and PFOS were quantified. Their occurrence cannot reliably distinguish AFFF from landfill leachate, wastewater, industrial discharge, atmospheric deposition, PFAS-treated consumer/textile materials, or other sources. Future source apportionment should include expanded target suites (e.g., PFHxS, PFBA, PFPeA, PFHxA, PFHpA, 6:2 FTS), TOP assay, suspect/non-target high-resolution mass spectrometry, and direct analysis of fire-fighting-foam products used in Ghana.

4.1.5 Lack of Seasonal Variation Assessment

Sampling was conducted once, from October to December 2022. Consequently, the dataset does not capture variability associated with wet/dry season, rainfall immediately before sampling, groundwater recharge, dilution, runoff pulses, or post-fire elapsed time. Apparent spatial differences may therefore partly reflect temporal conditions and should not be generalized to year-round concentrations.

4.1.6 Lack of Information on Foam

Incident records did not provide consistent information on fire magnitude, duration, fuel/material type, foam brand/formulation, volume applied, application duration, runoff containment, or post-fire cleanup. Thus, firefighting practices cannot be assumed to have been similar among sites, and the observed concentration differences cannot be quantitatively attributed to foam use.

4.1.7 Lack of Incorporating a Broader Suite of PFAS Analytes

The restricted analyte list is a major limitation. Although PFOA and PFOS remain environmentally

relevant legacy PFAAs, contemporary AFFF and other PFAS sources can contain short-chain PFAAs, fluorotelomer compounds, ether PFAS, and transformable precursors. Discussion of total PFAS burden, precursor transformation, and AFFF composition is therefore clearly separated from measured results. Follow-up work should use expanded targeted LC–MS/MS, TOP assay, suspect/non-target screening, seasonal monitoring, groundwater monitoring, sediment/biota analysis, dietary assessment, occupational exposure assessment, and human biomonitoring.

4.1.8 Limitation on HQ/HI

Deterministic HQ/HI values inherit analytical uncertainty and are also sensitive to exposure assumptions. The revised analysis uses final U.S. EPA 2024 RfDs and separates surface water from shallow groundwater, but it still assumes a single adult ingestion rate and body weight. Future risk assessment should apply probabilistic Monte Carlo methods, age- and sex-specific exposure factors, sensitive-population scenarios, cumulative exposure across a broader PFAS mixture, dietary pathways, and occupational exposure for firefighters and other highly exposed groups.

4.1.9 Budgetary Constraints

Budgetary and field constraints prevented measurement of soil organic carbon, texture, pH, bulk density, sediment characteristics, and site-specific hydrogeology. Because these variables strongly influence PFAA sorption and mobility, mechanistic interpretation is intentionally limited and should be addressed in future work.

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Data Availability Data for study will be made available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Conflict of interest The authors affirm that there was no possibility of a conflict of interest during the creation of this work.

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